



Degree Project in Computer Science and Engineering

Second cycle, 30 credits

Adversarial training for the automatic diagnosis of the electrocardiogram

Making deep learning analysis of the electrocardiogram
robust

ARVID ERIKSSON

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Degree Programme in Computer Science and Engineering

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Djupinlärning, Motstridig träning, Elektrokardiogram

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Acknowledgments

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I would like to thank xxxx for having yyyy. Or in the case of two authors: We would like to thank xxxx for having yyyy.

Stockholm, April 2026

Arvid Eriksson

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List of acronyms and abbreviations

APGD	Auto-PGD
CNN	convolutional neural network
ECG	electrocardiogram
FGSM	Fast Gradient Signed Method
PGD	projected gradient descent
SDG	Sustainable Development Goal
UN	United Nations

Chapter 1

Introduction

In this thesis, we examine the effects of adversarial attacks and training on deep learning models for **electrocardiogram (ECG)** analysis. Adversarial attacks are in this context defined as perturbations of model input that seek to alter the input by making an indistinguishable change in order to trick the model into making an incorrect classification or maximize the regression error. A model that can be tricked in such a manner is *vulnerable* to adversarial attacks, while a model resistant to such attacks, usually up to some perturbation threshold ϵ , is *robust*.

Adversarial training is in turn a method to make a model resistant to adversarial attacks. This is generally done by training on adversarial examples, that is input created by adversarial attacks. These concepts will be examined in this thesis from the context of **ECG** analysis using deep learning models.

This chapter describes the problem statement of the thesis, its context, the following goals, and the structure of the thesis.

1.1 Background

Cardiovascular diseases account for one-third of the deaths worldwide and the **ECG** is a major tool in their diagnoses. Deep neural networks have recently achieved striking success in diverse tasks such as image analysis and natural language processing, and there are great expectations when it comes to how this technology may improve health care and clinical practice. The last few years saw this technology present promising results in the interpretation of the **ECGs** for cardiovascular diagnosis and stratifying mortality risks and there are high expectations that these tools could enable the early and accurate detection of cardiac diseases [1, 2].

Due to the many interacting components and a large number of parameters, deep neural network models lack robustness meaning classification accuracy, and other key metrics, can often heavily decrease when small perturbations on the input are applied [3]. Models for medical applications need to work in scenarios that are potentially different than they were trained on are sought after. This is especially key as there are many different ECG datasets available internationally, where tools developed in one context might be relevant in others.

Adversarial training [4] is an effective approach for training deep learning models that are more robust to perturbations. Mathematically, adversarial training is formulated as a min-max problem, searching for the best solution to the worst-case attacks. It considers the situation where the training samples have been modified by an adversary. Since training samples are adversarially perturbed it is expected that the model will be more robust when faced with new perturbed samples, as well as generally noisy samples.

Another byproduct of adversarial training is that the adversarially trained model tends to produce better representations of the data which generally improves transferability [5]. In a medical setting, this is especially important as exact data distribution will differ between different settings, such as between different hospitals. Adversarial training could thus help provide models that represent data, such as ECGs samples, generally enough such that a model can easily be applied across different settings. Such a model would then help save resources as new settings and thus domain shifts would not require the training of new models.

1.2 Problem

The problem approached in this thesis is the adversarial vulnerability in state-of-the-art ECG deep learning models. The intention of this thesis is to promote more robust deep learning models through adversarial training, which can also be transferred to new settings.

1.2.1 Research question

What effect do adversarial attacks and training have on the classification accuracy of deep learning models used for the automatic diagnosis of abnormalities such as atrial fibrillation and first-degree atrioventricular block in the 12-lead ECG?

The research question is further split into the following sub-questions:

- How vulnerable are current state-of-the-art **ECG** diagnosis deep learning models to adversarial attacks, specifically gradient-based attacks such as **projected gradient descent (PGD)** and **Fast Gradient Signed Method (FGSM)**?
- How will adversarial training using **PGD** of state-of-the-art **ECG** models impact this vulnerability?
- How will adversarial training affect the transferability of a state-of-the-art **ECG** model to other **ECG** datasets that the model has not been trained on?

1.3 Purpose

The purpose of this project is to advance the field of deep learning analysis of **ECGs**. This will in turn help further sustainable health, give medical doctors tools for diagnosis assistance, and promote trust and robustness of used models.

A secondary purpose, as this is my master's thesis, is to attain a master's degree in Computer Science. This thesis as such thus seeks to prove that I fulfill the requirements for attaining a master's degree.

1.4 Goals

The research question is broken down into the following concrete goals:

- Measure the effects of adversarial attacks on state-of-the-art **ECGs** diagnosis models on key metrics such as accuracy and F-score.
- Measure how adversarial training of state-of-the-art **ECG** deep learning models improves their robustness to adversarial attacks and random noise.
- Measure how adversarial training affects the transferability of state-of-the-art **ECG** deep learning models to new datasets.

1.5 Research Methodology

This thesis will conduct a general empirical study by using public datasets for model training and evaluation, as is common in deep learning. The evaluation

will be done using test data from both the same datasets and different datasets than the ones trained on. The effects of adversarial attacks and training on the different generated models will thus be evaluated by obtaining metrics, such as accuracy, F1-score, and average precision from the predictions of the models on the test data. This will give an accurate estimate of the effects of adversarial attacks on the different models and allow us to estimate the effects of adversarial training. Bootstrapping will also be utilized to acquire an estimate of the confidence intervals of these metrics. The comparisons of the different models' performance will then be used to answer the research question with confidence.

1.6 Delimitations

The scope of the project is adversarial training and how this affects metrics such as accuracy and F1-score on different datasets, with and without adversarial attacks and random noise. Transferability to new datasets will also be investigated. Other related topics such as investigating novel types of models, such as non-convolutional neural networks, suitable for ECG data or the interpretability of the models will not be considered. Implementation of the resulting model in practice, e.g. in a real-life medical setting, is also not in the scope of the project.

Furthermore, this thesis will only investigate white-box adversarial attacks where full knowledge of the model parameters is known. This is due to the scope of robustness of the thesis over securing the model from a potential adversary. Adversarial training will also be the sole defense method for investigating, other defense methods such as defensive distillation will not be considered as they have been shown to have worse performance in several applications [6].

Chapter 2

Background

This chapter provides background information about deep learning and convolutional neural networks used for diagnosis of ECG abnormalities. Additionally, this chapter describes adversarial attacks and adversarial training. The chapter also describes related work.

2.1 Deep learning

In the past decade machine learning, and especially deep learning, has seen massive growth due to its performance gains in various fields. It has been used in data-driven decision-making in fields such as healthcare, finance, marketing, and education with great success and is beginning to take a central role in society [7].

Deep learning in turn consists of artificial neural networks which are compositions of functions applied to sets of computational nodes given some parameters. These functions are applied set-wise where such a set is called a layer and where the output of a layer is the input for the function of the next layer. See Figure 2.1 for an example artificial neural network where each layer is the result of a function, usually a composition of a non-linear and linear function, applied on the previous layer.

Deep learning, which is generally seen as artificial neural networks with multiple layers, contains many different model architectures that are useful for different scenarios. These architectures give a framework for the relationships between each layer and what types of functions are used. For example, the transformer architecture [8] is very efficient in natural language processing while the convolutional neural network architecture is efficient in image analysis due to its translational invariance property [9].

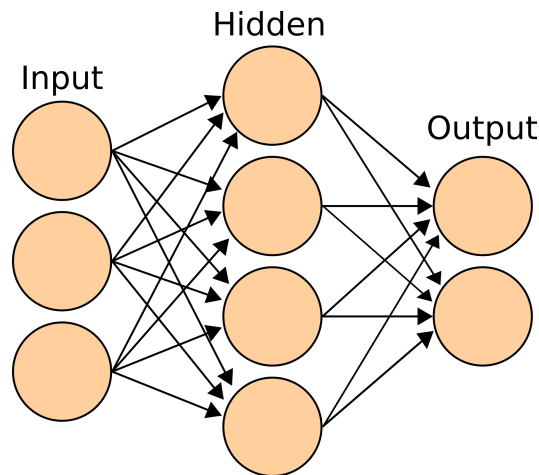


Figure 2.1: An artificial neural network with a hidden layer, courtesy of Colin M.L. Burnett.

2.1.1 Deep learning for the electrocardiogram

Artificial intelligence in general and deep learning especially has seen great success in the automatic analysis of the ECG [10]. Deep learning has, for example, been shown to be able to with great accuracy detect left ventricular dysfunction [11], atrial fibrillation [1], and hypertrophic cardiomyopathy [12] as well as predict other features such as age, which in turn can be a predictor for general health [13]. This makes deep learning models promising tools for medical professionals and an important topic for further advancement of ECG analysis.

The reason for deep learning's efficiency in the analysis of the ECG can be attributed to both an abundance of data and suitable model architectures. Since the ECG is a common recording there are several public datasets for such recordings, some of which are available publicly with more than 100 000 unique samples [14, 15]. The amount of data is a key factor for the performance of deep learning models and this relatively high amount of data therefore yields high-performing models.

As model architecture is also vital for the performance of the models another attributing factor is the recent use of convolutional neural networks using residual connections. Convolutional neural networks, which is a type of deep learning architecture, is a state-of-the-art architecture used in deep learning ECG analysis [10]. The suitability of this architecture can be motivated by its translational invariance property which, much like in

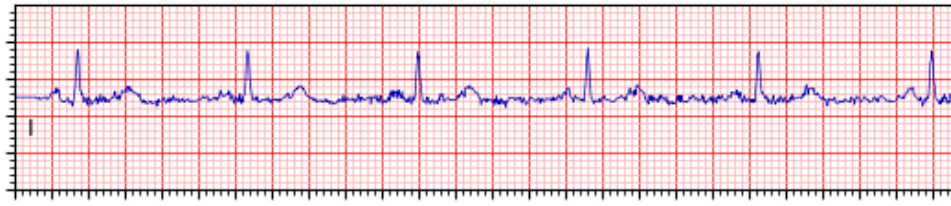


Figure 2.2: An example of an electrocardiogram from the PTB-XL dataset [15].

computer vision, allows the model to find patterns in the data regardless of where in the sequence it is located. This means that the model often will be able to capture important features, such as rapid and irregular beatings signifying atrial fibrillation, independent of where in the **ECG** these patterns can be observed.

2.2 Adversarial attacks

While deep learning models have shown striking performance in various number of areas they often also suffer from being sensitive to small perturbations, especially adversarially generated perturbations [16]. These adversarially generated perturbations, or adversarial attacks, are constructed to modify the input to a model by a small margin to create new input indistinguishable by a human from the original which makes the model make an erroneous prediction. Common examples of this are seen in computer vision, where an image of a certain class has its pixels modified by a very small amount to generate a nearly identical image that the model confidently sees as an entirely different class. For **ECGs**, such attacks can instead be seen as tiny changes in the signal, often with some type of smoothing to make the perturbations indiscernible from the original.

Adversarial attacks can generally be defined as a set of perturbation algorithms that seek to, given some knowledge of a model, create an indiscernible change in the data that makes the model misclassify the data. These algorithms can further be divided into black-box and white-box algorithms which assume limited or full knowledge of the model respectively. This thesis focuses on the robustness aspect of adversarial training rather than the security aspect and thus white-box algorithms will only be considered.

A common adversarial white-box attack is **FGSM** and its corresponding

generalization **PGD** (both described in detail in Section 3.2). These have seen wide use in the field of adversarial attacks due to their simplicity and efficiency in finding adversarial examples under some maximum perturbation ϵ [6].

2.2.1 Measuring robustness

As robustness to adversarial attacks often wants to be compared between models suitable metrics for measuring robustness are needed. Large studies have researched different metrics for robustness and their relevance for adversarial attacks [17]. Some such metrics search an entire region around each sample to find an accurate worst-case scenario, however, such methods are extremely costly. A good compromise between finding a worst-case scenario and being computationally feasible is to use the accuracy for adversarial samples generated by **PGD** for some range ϵ on test datasets from the same distribution as the one trained on [6]. However, this metric also depends on the parameters of the **PGD** which tend to differ across papers, making comparisons difficult. Efforts have been made to move to versions with perturbation range ϵ as the only parameter however these are only recently being adapted [18].

2.3 Adversarial training

Adversarial training is generally considered to be the most effective defense against adversarial attacks [19, 20, 21] and consists of including adversarial samples during training. Adversarial training thus tries to create a robust model that can classify even the worst perturbed samples (which are still within the true label's class) correctly. This can therefore be formulated as a min-max problem where the training procedure seeks to find the best solution for the worst-case attacks:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim D} \left[\max_{\delta \in S} L(\theta, x + \delta, y) \right] \quad (2.1)$$

where $(x, y) \sim D$ represents the data sampled from the dataset D and S is the set of allowed perturbations. S is usually dependent on some ϵ under the condition $\|\delta\|_p \leq \epsilon$, where p tends to be either 2, representing the euclidean norm, or ∞ , representing the max-norm.

In practice, adversarial training takes the form of using some type of perturbation algorithm to perturb the input samples during training. An example is adding **FGSM** to the loss function during training, although this

has been shown to sometimes lead to a failure mode called "catastrophic overfitting" [22]. While training using **PGD** is slower by a factor equal to the number of steps used, it is still seen as state-of-the-art in adversarial training. A problem with **PGD** is the need for hyperparameter tuning since **PGD** takes arguments such as the number of steps, step size, and projection size. There have been attempts to solve this, however, such as using a parameter-free adaptive tuning of the hyperparameters through **Auto-PGD (APGD)** introduced in [18].

2.4 Related work

This thesis is mainly based on two previous works, [1] which provides a deep learning model for automatic diagnosis of the 12-lead **ECG**, and [3] which shows that **ECGs** are susceptible to adversarial attacks.

Ribeiro et al. [1] shows how residual convolutional neural networks can be utilized for multilabel classification of 12-lead **ECGs** in order to diagnose several abnormalities such as atrial fibrillation and first-degree atrioventricular block. The work also produces several repositories containing code which is reused in this thesis, mostly for creating similar models.

Han et al. [3] provide an overview of how adversarial attacks can be applied to deep learning in **ECG** analysis. The work shows that deep learning models for **ECG** analysis are in general susceptible to adversarial attacks but that most adversarial attacks will create unnatural sequences that can be identified as artificial by a medical professional. They therefore also apply Gaussian smoothing operations on the adversarial examples which retain the examples' adversarial qualities while making the examples indistinguishable from their corresponding true samples. They therefore provide a clear framework for generating adversarial samples for **ECG** data.

The concept of adversarial training for **ECG** analysis has been studied before in [23, 24, 25] under various scopes. [23] investigates using a Conditional Generative Adversarial Network for adversarial training, while [24] attempts combining defensive distillation with adversarial training with promising results. [25] analyses how adversarial training and some other training regimes provide robustness to adversarial and random noise and implement this using a **convolutional neural network (CNN)** capable of variable input length. These papers, however, limit their experiments to training and testing on the same dataset and make no transferability experiments. They also base their experiments of robustness on **PGD** with different hyperparameters. A further potential avenue could therefore be the

parameter-free **APGD**.

Chapter 3

Methods

3.1 Data

This thesis utilizes two main datasets for its experiments: the CODE dataset [14] and PTB-XL dataset [15].

The CODE dataset consists of 2 322 513 12-lead **ECG** records from exams taken in Minas Gerais, Brazil, between 2010 and 2016 from 1 676 384 different patients. The dataset covers various abnormalities such as first-degree AV block, right bundle branch block, left bundle branch block, sinus bradycardia, atrial fibrillation, and sinus tachycardia. There is also a public dataset, CODE-15% containing 15% of the samples. This is the dataset used for this thesis.

PTB-XL is a dataset consisting of 21 799 12-lead **ECG** records collected between 1989 and 1996 from 18 869 different patients. It covers a larger range of diagnostic statements than the CODE dataset, with additional types of arrhythmias and diagnostics annotated.

3.2 Types of adversarial attacks

This section describes the adversarial attacks used in this thesis in detail.

3.2.1 Fast Gradient Signed Method

The **FGSM** is an adversarial attack that seeks to maximize the loss by taking a small step, ϵ , along the gradient of the loss function L . See Equation 3.1, where x is the input, y is the correct label, and θ is the model parameters.

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x L(\theta, x, y)) \quad (3.1)$$

3.2.2 Projected gradient descent

PGD is a generalization of **FGSM** where an arbitrary number of steps can be taken instead of **FGSM**'s single step. This is done by iteratively through gradient descent taking several steps in the gradient of the loss originating from the previous step. See Equation 3.2 for a formal definition for a sample x , using S as the boundary of the **PGD**, α as the step size, y as the correct label, and θ as the model parameters.

$$x^{t+1} = \Pi_{x+S}(x^t + \alpha \operatorname{sgn}(\nabla_x L(\theta, x^t, y))) \quad (3.2)$$

An important note here is that since a new loss is calculated each iteration, the final adversarial sample will be costly to generate, having the number of gradient calculations scaling linearly with the number of steps. Doing this for many samples will therefore have a high computational overhead. This is extremely expensive in computer vision, but the cost is manageable with **ECGs** due to the lower dimensionality. There are also other methods [26] that show similar performance to **PGD** with a computational cost close to **FGSM**, which are also worth considering if computational cost is a limiting factor.

3.3 Deep learning architectures

As this thesis seeks to improve state-of-the-art residual **CNNs** a detailed description of their main properties will be described here.

3.3.1 Convolutional neural networks

As convolutional neural networks are a central contributor to the high performance of deep learning models in **ECG** analysis, and thus a central part of this thesis, this section seeks to describe the architecture in enough detail that the main properties of a convolutional neural network can be understood.

Convolutional neural networks are a subset of artificial neural networks where some of the functions applied to layers perform convolutions. These convolutions generally take the form of filters that slide over an image or a time series and create an output that is the input multiplied by the kernel as it slides over the input. See Figure 3.1 for a one-dimensional example similar to the case for an **ECG**. These kernels are usually multiple, which yields a multi-dimensional output which is then aggregated.

As kernel parameters are the only weights being modified in the layer, and as this kernel is generally quite small, there is a significant reduction in the

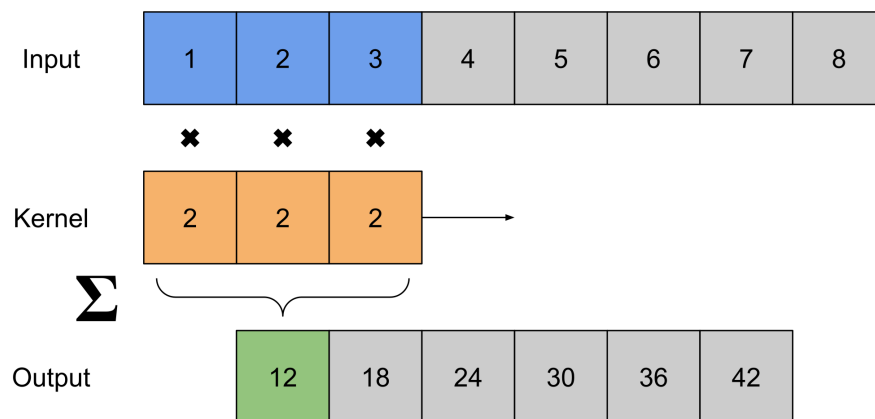


Figure 3.1: Example of a one-dimensional convolutional layer without a bias. The kernel contains the trainable parameters.

number of parameters being optimized compared to a fully connected layer. In Figure 3.1 a corresponding fully connected layer would have $8 \times 6 = 48$ weights compared to the 3 used in the convolutional case. This reduction in parameters leads to faster training while introducing a certain bias to the model where spatial features are sought after, thus making the model ideal for computer vision and time series analysis where a pattern at an unknown location in the sequence is sought after, as for ECG analysis. These layers are generally also combined with pooling layers and residual connections which help aggregate and speed up the training of the model.

Chapter 4

Implementation

[Vad gjorde du? Hur gick det till? – Välj lämplig rubrik
("Genomförande", "Konstruktion", "Utveckling" eller annat)]

What have you done? How did you do it? What design decisions did you make? How did what you did help you to meet your goals?

Vad du har gjort? Hur gjorde du det? Vilka designval gjorde du?
Hur kom det du hjälpte dig att uppnå dina mål?

4.1 Hardware/Software design .../Model/Simulation model & parameters/...

Hårdvara / Mjukvarudesign ... / modell / Simuleringsmodell och parametrar / ...

Figure 4.1 shows a simple icon for a home page. The time to access this page when served will be quantified in a series of experiments. The configurations that have been tested in the test bed are listed in Table 4.1. In 7.0 % of cases, there was an error indicating xxxxx.

Figur 4.1 visar en enkel ikon för en hemsida. Tiden för att få tillgång till den här sidan när den laddas kommer att kvantifieras i en serie experiment. De konfigurationer som har testats i provbänk listas i tabell 4.1.

Vad du har gjort? Hur gjorde du det? Vilka designval gjorde du?

Testade konfigurationer

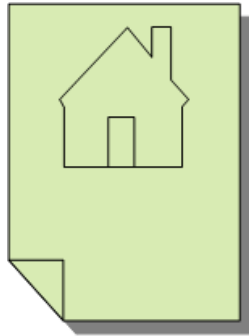


Figure 4.1: Homepage icon

Table 4.1: Configurations tested

Configuration	Description
1	Simple test with one server
2	Simple test with one server

4.2 Implementation .../Modeling/Simulation/...

Implementering ... / modellering / simulering / ...

Two commonly used simulators are:

- Mininet** This simulator uses traffic control (tc) to simulate network devices connected by links with specific bandwidth, packet loss rates, qdisc methods, etc.
- ns-2 or ns-3 simulator** These simulators are very useful for simulating wireless communication links between moving devices. You can specify the mobility patterns of the nodes.

4.2.1 Some examples of coding

This section is simply to show some example of how you can include code in your thesis - this is not a section you would have in your thesis.

Det här avsnittet är helt enkelt för att visa ett exempel på hur du kan inkludera kod i ditt examensarbete - det här är inte ett avsnitt du skulle ha i ditt examensarbete.

Listing 4.1 shows an example of a simple program written in C code.

Listing 4.1: Hello world in C code

```
int main() {
    printf("hello ,\uworld");
    return 0;
}
```

In contrast, Listing 4.2 is an example of code in Python to get a list of all of the programs at KTH.

Listing 4.2: Using a python program to access the KTH API to get all of the programs at KTH

```
KOPPSbaseUrl = 'https://www.kth.se '
```

```
def v1_get_programmes():
    global Verbose_Flag
    #
    # Use the KOPPS API to get the data
    # note that this returns XML
    url = "{0}/api/kopps/v1/programme".format(KOPPSbaseUrl)
    if Verbose_Flag:
        print("url:\u" + url)
    #
    r = requests.get(url)
    if Verbose_Flag:
        print("result\uof\ugetting\uv1\uprogramme:\u{}".format(r.text))
    #
    if r.status_code == requests.codes.ok:
        return r.text                # simply return the XML
    #
    return None
```

4.2.2 Some examples of figures in tikz

This section is simply to show some example of how you can draw your own figures for in your thesis - this is not a section you would have in your thesis.

Det här avsnittet är helt enkelt för att visa ett exempel på hur du kan rita dina egna figurer i ditt examensarbete – det här är inte ett avsnitt du skulle ha i ditt examensarbete.

These figures are just some examples to show that you can draw your own figures for in your thesis. This has two advantages: *(i)* you do not have to worry about copyrights – as these are your own figures and *(ii)* the text is now readable and not simply a picture of text – so screen readers can read the figure's contents to someone who is listening to the contents of your thesis.

4.2.2.1 Azure's Form Recognizer

Figure 4.2 shows the processing of key-value extraction from a PDF document using Azure's Form Recognizer.

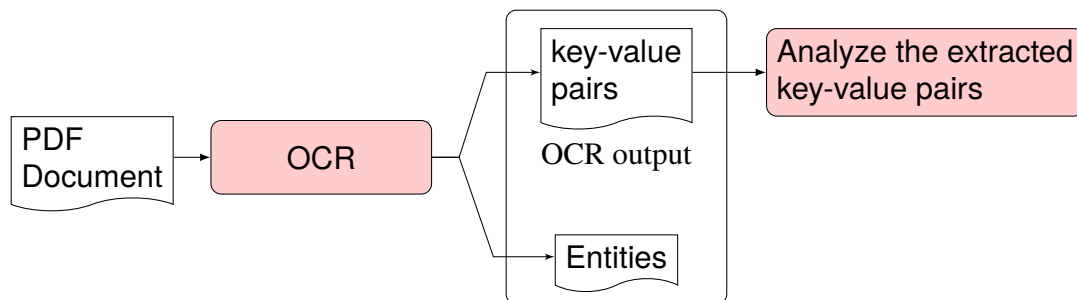


Figure 4.2: The processing of key-value extraction from a PDF document using Azure's Form Recognizer

4.2.2.2 Hyper-V with Containers

Figure 4.3 shows how Hyper-V deals with containers.

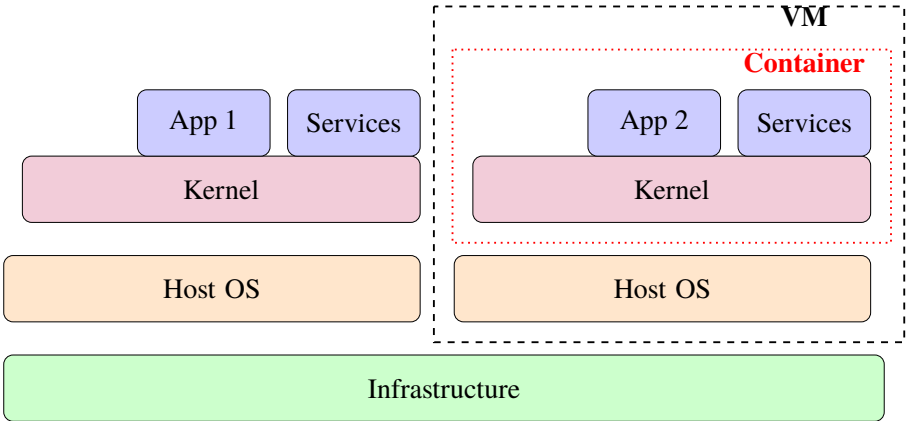


Figure 4.3: Hyper-V with containers

4.2.2.3 VM versus Containers

Figure 4.4 shows a comparison of virtual machines (VMs) versus containers.

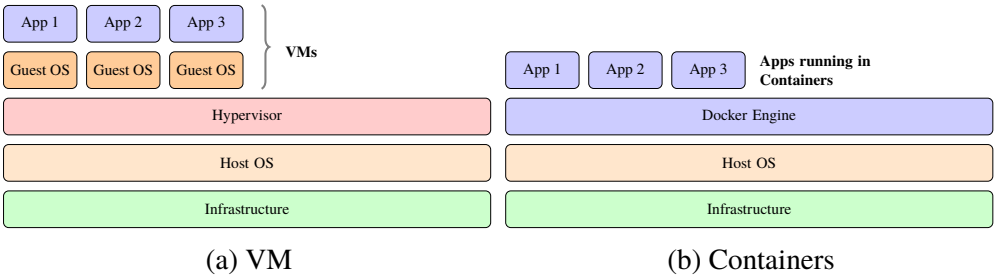


Figure 4.4: Virtual machines (VMs) versus Containers

Chapter 5

Results and Analysis

svensk: Resultat och Analys

Sometimes this is split into two chapters.

Keep in mind: How you are going to evaluate what you have done?

What are your metrics?

Analysis of your data and proposed solution

Does this meet the goals which you had when you started?

In this chapter, we present the results and discuss them.

I detta kapitel presenterar vi resultaten och diskutera dem.

Ibland delas detta upp i två kapitel.

Hur du ska utvärdera vad du har gjort? Vad är din statistik?

Analys av data och föreslagen lösning

Innebär detta att uppfyllelse av de mål som du hade när du började?

5.1 Major results

Huvudsakliga resultat

Some statistics of the delay measurements are shown in Table 5.1. The delay has been computed from the time the GET request is received until the response is sent.

Lite statistik av fördröjningsmätningarna visas i Tabell 5.1. Förseningen har beräknats från den tidpunkt då begäran GET tas emot fram till svaret skickas.

Table 5.1: Delay measurement statistics

Configuration	Average delay (ns)	Median delay (ns)
1	467.35	450.10
2	1 687.5	901.23

Table 5.2 shows the measurement of round trip times from four hosts to and from a server.

Table 5.2: Result for the ping measurements of RTT for 4 hosts

Host	host to server RTT in ms			
	min	avg	max	mdev
h1	5.625	5.625	5.625	0.0
h2	2.909	2.909	1.909	0.0
h3	5.007	5.007	5.007	0.0
h4	2.308	2.308	2.308	0.0

Fördröj mätstatistik

Konfiguration | Genomsnittlig fördröjning (ns) | Median fördröjning (ns)

Figure 5.1 shows an example of the performance as measured in the experiments.

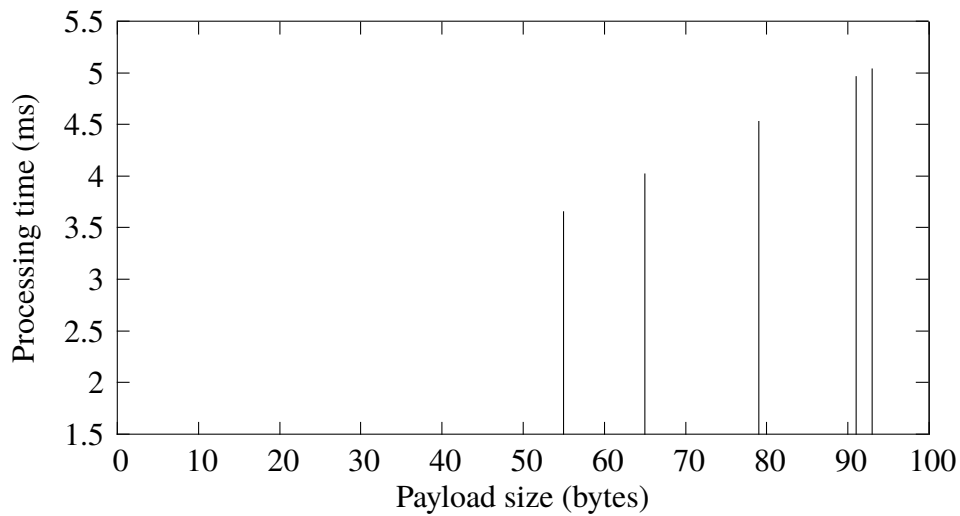


Figure 5.1: Processing time vs. payload length

Given these measurements, we can calculate our processing bit rate as the inverse of the time it takes to process an additional byte divided by 8 bits per byte:

$$\text{bit rate} = \frac{1}{\frac{\text{time}_{\text{byte}}}{8}} = 20.03 \text{ kb/s}$$

Table 5.3 shows another table in which some values have been set in bold (using \B) to emphasize them. Note how the S formatting has been modified so that it considers the weight of the characters and this is able to decimal align even these hold-faced numbers with the numbers in the column above them.

Table 5.3: Median values of sandwich attributes

Attribute	sites	
	A	B
price (in SEK)	36.5	71.3
protean (g)	97.2	100.0
salt (mg)	9.7	9.3
Average customer rating in %	82.2	89.9

Figure 5.2 shows a stacked bar chart using pgfplots. It illustrates how easy it is to take a set of data and make a stacked bar plot. One of the features is the shifted values – this is very useful when the bar itself is too small to put the value into.

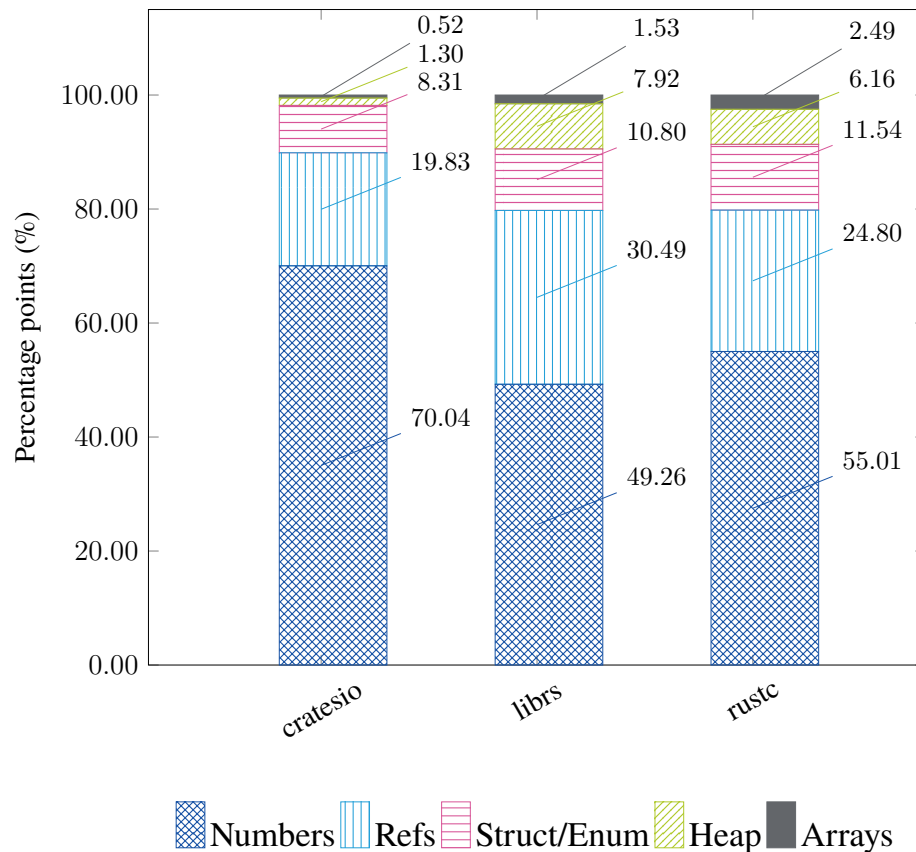


Figure 5.2: Rust types distribution for the compiler, crates.io, and lib.rs. (percentage) - appears here with the permission of the author - see the thesis at <https://urn.kb.se/resolve?urn=urn%3Anbn%3Ase%3Akh%3Adiva-332124>

5.2 Reliability Analysis

Analys av tillförlitlighet
Tillförlitlighet i metod och data

5.3 Validity Analysis

Analys av validitet
Validitet i metod och data

Chapter 6

Discussion

Diskussion
Förbättringsförslag?

This can be a separate chapter or a section in the previous chapter.

Chapter 7

Conclusions and Future work

Slutsats och framtida arbete

Add text to introduce the subsections of this chapter.

7.1 Conclusions

Slutsatser

Describe the conclusions (reflect on the whole introduction given in Chapter 1).

Discuss the positive effects and the drawbacks.

Describe the evaluation of the results of the degree project.

Did you meet your goals?

What insights have you gained?

What suggestions can you give to others working in this area?

If you had it to do again, what would you have done differently?

Uppfyllde du dina mål?

Vilka insikter har du fått?

Vilka förslag kan du ge till andra som arbetar inom detta område? Om du skulle göra detta igen, vad skulle du ha gjort annorlunda?

7.2 Limitations

Begränsande faktorer

Vad gjorde du som begränsade dina ansträngningar? Vilka är begränsningarna i dina resultat?

What did you find that limited your efforts? What are the limitations of your results?

7.3 Future work

Vad du har kvar ogjort?

Vad är nästa självklara saker som ska göras?

Vad tips kan du ge till nästa person som kommer att följa upp på ditt arbete?

Describe valid future work that you or someone else could or should do. Consider: What you have left undone? What are the next obvious things to be done? What hints can you give to the next person who is going to follow up on your work?

Due to the breadth of the problem, only some of the initial goals have been met. In these section we will focus on some of the remaining issues that should be addressed in future work. ...

7.3.1 What has been left undone?

The prototype does not address the third requirement, *i.e.*, a yearly unavailability of less than 3 minutes; this remains an open problem. ...

7.3.1.1 Cost analysis

Example of a missing component

The current prototype works, but the performance from a cost perspective makes this an impractical solution. Future work must reduce the cost of this solution; to do so, a cost analysis needs to first be done. ...

7.3.1.2 Security

Example of a missing component

A future research effort is needed to address the security holes that results from using a self-signed certificate. Page filling text mass. Page filling text mass. ...

7.3.2 Next obvious things to be done

In particular, the author of this thesis wishes to point out xxxxxx remains as a problem to be solved. Solving this problem is the next thing that should be done. ...

7.4 Reflections

Reflektioner

Vilka är de relevanta ekonomiska, sociala, miljömässiga och etiska aspekter av ditt arbete?

What are the relevant economic, social, environmental, and ethical aspects of your work?

One of the most important results is the reduction in the amount of energy required to process each packet while at the same time reducing the time required to process each packet.

The thesis contributes to the **United Nations (UN) Sustainable Development Goals (SDGs)** numbers 1 and 9 by xxxx.

In the references, let Zotero or other tool fill this in for you. I suggest an extended version of the IEEE style, to include URLs, DOIs, ISBNs, etc., to make it easier for your reader to find them. This will make life easier for your opponents and examiner.

IEEE Editorial Style Manual: https://www.ieee.org/content/dam/ieee-org/ieee/web/org/conferences/style_references_manual.pdf

Låt Zotero eller annat verktyg fylla i det här för dig. Jag föreslår en utökad version av IEEE stil - att inkludera webbadresser, DOI, ISBN osv. - för att göra det lättare för läsaren att hitta dem. Detta kommer att göra livet lättare för dina opponenter och examinerator.

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Appendix A

Supporting materials

Here is a place to add supporting material that can help others build upon your work. You can include files as attachments to the PDF file or indirectly via URLs. Alternatively, consider adding supporting material uploaded as separate files in DiVA.

The BibTeX references used in this thesis are attached. 

Some source code relevant to this project can be found at <https://github.com/ggmaquirejr/E-learning> and <https://github.com/ggmaquirejr/Canvas-tools>.

Your reader can access the attached (embedded) files using a PDF tool such as Adobe Acrobat Reader using the paperclip icon in the left menu, as shown in Figure A.1 or by right-clicking on the push-pin icon in the PDF file and then using the menu to save the embedded file as shown in Figure A.2.

An argument for including supporting material in the PDF file is that it will be available to anyone who has a copy of the PDF file. As a result, they do not have to look elsewhere for this material. This comes at the cost of a larger PDF file. However, the embedded files are encoded into a compressed stream within the PDF file; thus, reducing the number of additional bytes. For example, the references.bib file that was used in this example is 10 617 B in size but only occupies 4 261 B in the PDF file.

DiVA is limited to ≈ 1 GB for each supporting file. If you have very large amounts of supporting material, you will probably want to use one of the data repositories. For additional help about this, contact KTH Library via researchdata@kth.se.

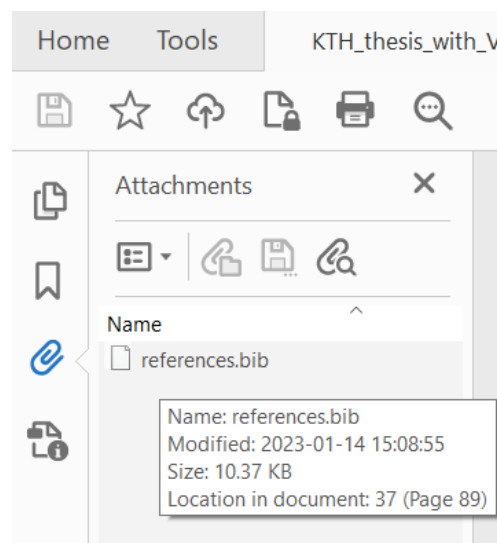


Figure A.1: Adobe Acrobat Reader using the paperclip icon for the attached references.bib file

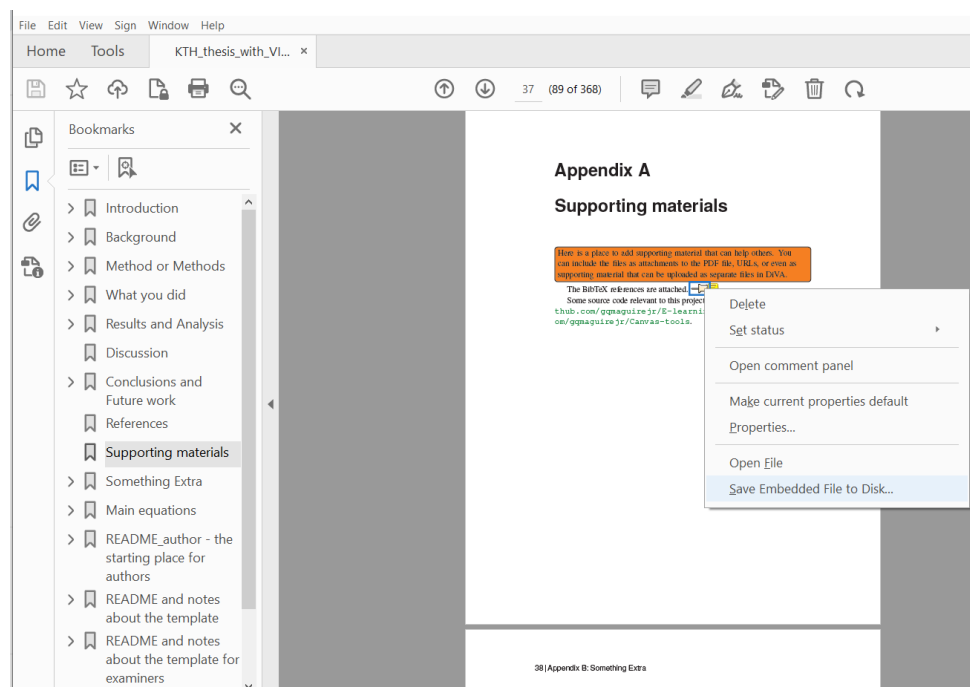


Figure A.2: Adobe Acrobat Reader after right-clicking on the push-pin icon for the attached references.bib file

Appendix B

Something Extra

svensk: Extra Material som Bilaga

B.1 Just for testing KTH colors

You have selected to optimize for print output

- Primary color

- kth-blue 

- kth-blue80 

- Secondary colors

- kth-lightblue 

- kth-lightred 

- kth-lightred80 

- kth-lightgreen 

- kth-coolgray 

- kth-coolgray80 

black 

Appendix C

README_author - the starting place for authors

This document, written by Gerald Q. Maguire Jr, describes the thesis template that I have developed for use at KTH Royal Institute of Technology (KTH). It is important to note that the template is **not prescriptive**, as not every thesis will have all of the parts that the template shows. However, if there is something that you decide to leave out, you should make a conscious decision to do so and you should consider the impact this may have on your thesis being approved by the examiner.

Fundamental to the design of the template are several key factors:

- Helping students be successful in their degree project,
- Helping students produce a high-quality thesis, and
- Supporting all of the (relevant) phases of the degree project process.

This document is a work in progress.

C.1 Advice for Author or Authors

One of the hardest problems an author faces is getting started writing, *i.e.*, the blank sheet of paper – empty file barrier. The template provides a non-blank starting point; hence, avoiding the blank paper barrier. Additionally, the template provides some initial structure, basically, an Introduction, Methods, Results, and Discussion (IMRAD) structure, so that there are hints of where to place material. Moreover, there are places (and notes) about material that the

student should consider adding; for example, the “required reflections” section in the final chapter.

The template (located in the file `examplethesis.tex`) also provides some examples of commonly occurring types of content, so that one can easily find examples of how to include a figure, table, code listing, *etc.* These examples are not meant to be exhaustive and quite often the student will probably need to learn new $\text{\LaTeX}$ commands in the course of writing their thesis.

As an author, the first step is to configure the $\text{\LaTeX}$ engine that you will use to process the files - see Appendix C.2. The second step will be to configure the template - see Appendix C.3. The third step will be to make sure that the information about you, your supervisor(s), and the examiner are correct in the file `custom_configuration.tex` - this information uses the macros described in Appendix C.4. Now that you have a lot of the administrative details taken care of it is time to start to write - see Appendix C.5.

Note that if you are using Overleaf:

Make your own copy of the template If you have opened the template from a URL, in the upper left-hand corner, click on **Menu**. Then select **Copy Project** - this will give you your own private copy.

Use a helpful project name I suggest you include your name in the project name so that when you share it with your supervisor(s) and examiner, they will know it is your project.

Invite your supervisor(s) and examiner to your project You can invite your supervisor(s) and examiner to your project and they can directly comment on and correct your drafts.

Log in to Overleaf with your KTH account If you log in to Overleaf with your KTH account, you get a version of Overleaf that lets you turn on “Track changes” which is very useful (particularly if you have invited your supervisor(s) and examiner to join your project). It also gives you a bit more of a time budget to compile (which can be useful if you have a lot of Tikz figures or other things that take a lot of time for the $\text{\LaTeX}$ engine to render).

If you have more detailed questions about the template itself - see Appendix D.

C.2 Author configuration of the $\text{\LaTeX}$ engine

The template should work with $\text{\pdfLaTeX}$, $\text{\XeLaTeX}$, and $\text{\LuaLaTeX}$. If you are using Overleaf, I strongly recommend using $\text{\XeLaTeX}$ — as this will get the `Arial` fonts correct for the KTH cover. If you are running the compiler on your local machine and you use $\text{\XeLaTeX}$ **and** you have `Arial` as a system font, then it will be able to use it. Similarly, for $\text{\LuaLaTeX}$. For $\text{\pdfLaTeX}$ I have used `\fontfamilyhelvet`, *i.e.*, Helvetica, as it is a sans serif font.

One student reported problems with `FONTSPEC` not loading the fonts properly when running locally with macOS 12.4, TeXLive 2022, LaTeX Workshop on VS Code, and $\text{\XeLaTeX}$ - the solution is described at <https://tug.org/TUGboat/tb39-2/tb122robertson-fontspec.pdf>.

If you are using Overleaf, it is easy to select the compiler (*i.e.*, $\text{\TeX}$ engine) by using the drop-down menu, as shown in Figure C.1.

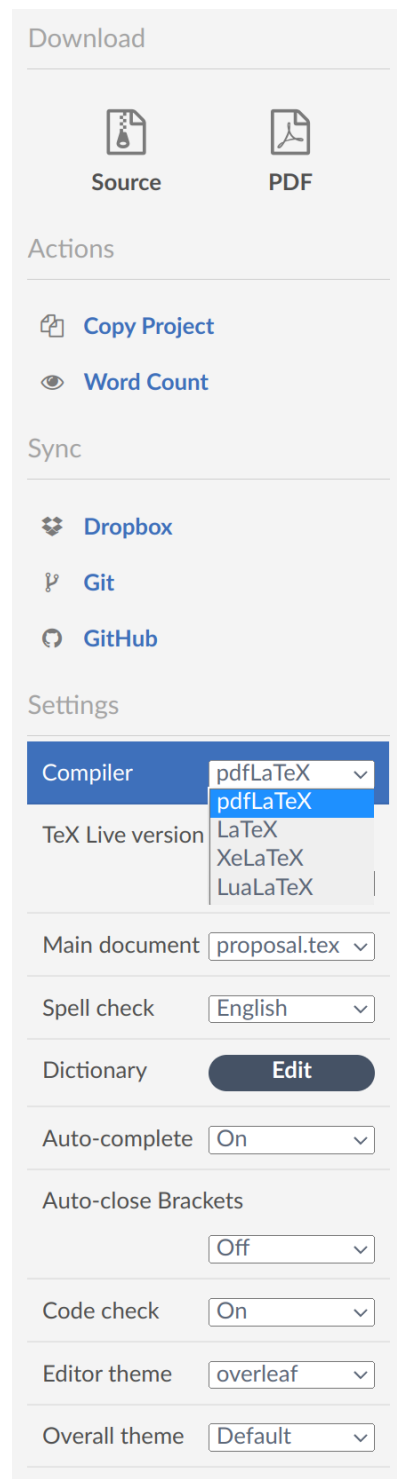


Figure C.1: Selecting a compiler (*i.e.*, TeX engine) in Overleaf

C.3 Author configuration of the template

The template is designed to handle a thesis written in English or Swedish. You can set the default language to ‘english’ or ‘swedish’ by passing an option to the documentclass. Note that the language option is written in all lowercase letters; for example, to set the document’s language to English:

```
\documentclass[english]{kththesis}
```

To set the document’s language to Swedish (uncomment the following line):

```
\documentclass[swedish]{kththesis}
```

The language option ‘swedish’ sets the conditional `\ifinswedish` to true. Among many other things, this conditional is used to configure the KTH cover and the title page to use the chosen language.

The two most common bibliographic engines are supported, *i.e.*, BibTeX and BibLaTeX. To set the language to English and use the bibliographic engine to BibTeX you would say:

```
\documentclass[english, bibtex]{kththesis}
```

To set the language to Swedish and use the bibliographic engine to BibLaTeX you would say:

```
\documentclass[swedish, biblatex]{kththesis}
```

The above illustrates that you can pass multiple options to the document class separated by commas. Also, note that the options were passed as all lowercase letters.

You can, of course, also modify the formatting of the citations and bibliography. See for example the following code snippet:

```
\ifbiblatex
% \usepackage[language=english, bibstyle=authoryear,
% ↪ citestyle=authoryear, maxbibnames=99]{biblatex}
% \usepackage[style=numeric, sorting=none, backend=biber
% ↪ ]{biblatex}
\usepackage[bibstyle=authoryear, citestyle=authoryear,
% ↪ maxbibnames=99, language=english]{biblatex}
% alternatively you might use another style, such as
% ↪ IEEE
% \usepackage[style=ieee]{biblatex}
\addbibresource{references.bib}
% \DeclareLanguageMapping{norsk}{norwegian}
```

```
\else
  % The line(s) below are for BibTeX
  \bibliographystyle{bibstyle/myIEEETran}
  %\bibliographystyle{apalike}
\fi
```

To optimize for digital output (this changes the color palette) add the option: `digitaloutput`. There are also options for A4 or G6 paper: `a4paper` or `g5paper` (respectively). There is an option for nomenclature, to produce and refer to equations. Finally, there are options for a 1st cycle thesis or 2nd cycle thesis: `bachelor` and `master` (respectively); however, these two options are **not** currently used.

One of the first things that the author(s) will want to do is add the working title and subtitle to the thesis. This is done using the `\title`, `\subtitle`, `\alttitle`, and `\altsubtitle` macros as shown below:

```
\title{This is the title in the language of the thesis}
\subtitle{A subtitle in the language of the thesis}

% give the alternative title - i.e., if the thesis is in
  ↪ English,
% then give a Swedish title
\alttitle{Detta är den svenska översättningen av titeln}
\altsubtitle{Detta är den svenska översättningen av
  ↪ undertiteln}
% alternative, if the thesis is in Swedish, then give an
  ↪ English title
%\alttitle{This is the English translation of the title}
%\altsubtitle{This is the English translation of the
  ↪ subtitle}
```

Setting these values once and then using them in many places reduces the work to change them while at the same time ensuring consistency.

Some additional configuration that the author(s) might do is to set the values of the macros related to the course cycle, course code, date of the thesis, number of credits, degree/exam name, subject area, and if the degree is done external to KTH to set the host information. Consider the snippet below for a student admitted to the “Bachelor’s Programme in Information and Communication Technology (TCOMK)” program and enrolled in the degree project course “IA150X Degree Project in Information and Communication Technology, First Cycle 15.0 credits” and working at a company “Företaget AB”:

```

\hostcompany{Företaget AB} % Remove this line if the
    ↪ project was not done at a host company

\date{\today}

\courseCycle{1}
\courseCode{IA150X}
\courseCredits{15.0}

\programcode{TCOMK}
\degreeName{Bachelors degree}
% Note that the subject area for a Bachelor's thesis (
    ↪ Kandidatexamen)
% should be either Technology or Architecture
% If the thesis is in Swedish, these would be: teknik |
    ↪ arkitektur
% -- Note the use of lower case for the Swedish subject
    ↪ area
\subjectArea{Technology}

```

Note that in the above macros you have to give the English or Swedish names in the arguments to \degreeName and \subjectArea - as shown below:

```

\degreeName{Kandidatexamen}
\subjectArea{teknik}

```

For a CDATE student enrolled in the course “DA231X Degree Project in Computer Science and Engineering, Second Cycle 30.0 credits”, the cycle, program, course code, degree, and subject area information would be:

```

\programcode{CDATE}
\courseCycle{2}
\courseCode{DA231X}
\courseCredits{30.0}
\degreeName{Degree of Master of Science in Engineering}
\subjectArea{Computer Science and Engineering}

```

The set of possible values for the English or Swedish names in the arguments to \degreeName are:

```

\degreeName{Higher Education Diploma}
\degreeName{Högskoleexamen}

\degreeName{Bachelors degree}
\degreeName{Kandidatexamen}

```

```

\degreeName{Master of Architecture}
\degreeName{Arkitektexamen}

\degreeName{Degree of Master of Science in Engineering}
\degreeName{Civilingenjörs}

\degreeName{Magister}
\degreeName{Magisterexamen}

\degreeName{Degree of Master of Science}
\degreeName{Masterexamen}

\degreeName{Master of Science in Engineering and Master
  ↪ of Arts in Education degree}
\degreeName{Civilingenjör och lärare examen}

\degreeName{Degree of Master of Science in Secondary
  ↪ Education}
\degreeName{Ämneslärarexamen}

\degreeName{Both} # Degree Project in the Field of
  ↪ Technology <teknikområde> and the Main Field of
  ↪ Study <huvudområde>
\degreeName{Same} # The case when the field of
  ↪ technology <teknikområde> and main field of study <
  ↪ huvudområde> are the same.

```

For the last two cases, the code compares the values of subjectArea and secondSubjectArea.

You can find a list of the program codes and school acronyms in the file: lib/schools_and_programs.ins.

There are a set of rules about what is to be displayed on the KTH cover. These can be found at <https://www.kth.se/social/group/sprakkommitte/page/omrade-for-examensarbete/>.

One of the reasons for many of the macros shown above and below is to collect the information that is needed to report the approved thesis in Digitala Vetenskapliga Arkivet (DiVA) and to report the title(s) and grade in Lokalt adb-baserat dokumentationssystem (LADOK).

National subject categories are a **required** field in the DiVA record. These categories follow a definition by SCB (nowadays known as Statistikmyndig-

heten or in English: Statistics Sweden) and HSV (Högskoleverket - nowadays known as Universitetskanslersämbetet (UK-ämbetet) and Universitets- och högskolerådet (UHR) or in English: Swedish Higher Education Authority and Swedish Council for Higher Education). While these codes refer to research areas, these codes are also used in KTH to indicate the area of the thesis. The guidance that I received from the Linköping University library was that one should try to use 5-digit codes when possible. Some examples of these codes are shown in Table C.1.

\nationalsubjectcategories{} comma separated list of national subject category codes - each a 3 or 5 digit code

An example for a thesis in Computer Science and Computer Systems:

```
\nationalsubjectcategories{10201, 10206}
```

You can find the subjects and their codes in:

<https://www.scb.se/contentassets/3a12f556522d4bdc887c4838a37c7ec7/standard-for-svensk-indelning--av-forskningsamnen-2011-uppdaterad-aug-2016.pdf>

and

<https://www.scb.se/contentassets/10054f2ef27c437884e8cde0d38b9cc4/oversattningsnyckel-forskningsamnen.pdf>

Table C.1: Examples of some national subject categories and their codes

Code	Category (in Swedish)	Category (in English)
102	Data- och informationsvetenskap (Datateknik)	Computer and Information Sciences
10201	Datavetenskap (datalogi)	Computer Sciences
10202	Systemvetenskap, informationssystem och informatik (samhällsvetenskaplig inriktning under 50804)	Information Systems (Social aspects to be 50804)
10203	Bioinformatik (beräkningsbiologi) (tillämpningar under 10610)	Bioinformatics (Computational Biology) (applications to be 10610)
10204	Människa-datorinteraktion (interaktionsdesign) (Samhällsvetenskapliga aspekter under 50803)	Human Computer Interaction (Social aspects to be 50803)
10205	Programvaruteknik	Software Engineering
10206	Datorteknik	Computer Engineering
10207	Datorseende och robotik (autonoma system)	Computer Vision and Robotics (Autonomous Systems)
10208	Språkteknologi (språkvetenskaplig databehandling)	Language Technology (Computational Linguistics)
10209	Medieteknik	Media and Communication Technology
10299	Annan data- och informationsvetenskap	Other Computer and Information Science
202	Elektroteknik och elektronik	Electrical Engineering, Electronic Engineering, Information Engineering
20201	Robotteknik och automation	Robotics
20202	Reglerteknik	Control Engineering
20203	Kommunikationssystem	Communication Systems
20204	Telekommunikation	Telecommunications
20205	Signalbehandling	Signal Processing
20206	Datorsystem	Computer Systems
20207	Inbäddad systemteknik	Embedded Systems
20299	Annan elektroteknik och elektronik	Other Electrical Engineering, Electronic Engineering, Information Engineering

C.4 Author macros

It is assumed that there can only be 1 or 2 authors. For many years now 2nd cycle theses are expected to only have one author.

For the author or first author, there are a number of macros defined to store information about the author, so that it can later be used in multiple places – for example, the KTH cover (produced with `\kthcover`), the title page (produced with `\titlepage`, the “For DiVA” section at the end of the thesis (produced with `\divainfo{pg:lastPageofPreface}{pg:lastPageofMainmatter}`), and possibly a JavaScript Object Notation (JSON) file named `fordiva.json` produced as a by product of the `\divainfo`. Note that the actual section name has DiVA set in all caps - which hopefully should not occur in the thesis! If the string DiVA set in all caps, does have to appear, then the section heading should be preceded by four euro signs and followed by four more euro signs (as is done this document).

The author-related macros are:

<code>\authorsLastname{}</code>	the last name of the author*
<code>\authorsFirstname{}</code>	the first name of the author
<code>\email{}</code>	the KTH e-mail address of the author
<code>\kthid{}</code>	the author’s kthid, this generally starts with the string “u1” and is a unique identifier for every KTH user.
<code>\authorsSchool{}</code>	the value is generally of the form: <code>\schoolAcronym{EECS}</code> . The currently supported school acronyms are: ABE, CBH, EECS, ITM, and SCI. These are defined in the file <code>schools_and_programs.ins</code> .

If the first author is not in Stockholm, Sweden when the acknowledgements are written, then add that information via the macros described below. This information will be used when generating the acknowledgements signature. The acknowledgements signature is the text at the end of the

*Note that the author’s name can include a suffix such as “, Jr.” or “ Jr.”, i.e., the suffix can be separated with a comma or not – as the author prefers to write their name.

acknowledgements and it gives the place where the author(s) is/are when writing the acknowledgements and also gives the date and name(s).

`\authorCity{A City}` specify the city

`\authorCountry{A Country}` specify the country

`\authorCityCountryDate{}` pass into this function the month and year for the acknowledgement. This can be a string such as January 2022 or it can be a $\text{\LaTeX}$ expression, such as `\MONTH\enspace\the\year`.

If there is a second author and the place, month, and year are **all** the same, then specify the month and year for only the **first** author:

```
\authorCityCountryDate{\MONTH\enspace\the\year}
```

If there is a second author and the place is different, then say:

```
\authorCityCountryDate{ }
```

If there is a second author, the macros are:

`\secondAuthorsLastname{}` the last name of the 2nd author
`\secondAuthorsFirstname{}` the first name of the 2nd author
`\secondemail{}` the KTH e-mail address of the 2nd author
`\secondkthid{}` the 2nd author's kthid
`\secondAuthorsSchool{}` the school of the 2nd author

If the second author is not in the same place as the first author, then add the relevant information using the macros below. This information will be used when generating the acknowledgements signature.

`\secondAuthorCity{A City}` specify the city
`\secondAuthorCountry{A Country}` specify the country
`\secondAuthorCityCountryDate{\MONTH\enspace\the\year}`
 pass into this function the month and year for the acknowledgement

If the second author is the same place as the first author, then comment out or delete the `\secondAuthorCityCountryDate{}` as shown below:

```
%\secondAuthorCityCountryDate{}
```

C.5 Starting to write

As you write you will notice "todo" notes in the template. They follow the following conventions:

```
\generalExpl{Comments/directions/... in English}
\sweExpl{Text på svenska}
\engExpl{English descriptions about formatting}
\sweExpl{warnings}
```

C.5.1 Working abstract

I generally recommend that every student start by writing a working abstract, this will help you keep your focus. To find where you can start to enter your abstract look in the *examplethesis.tex* file for the line:

```
\generalExpl{Enter your abstract here!}
```

There is lots of information already in the template to help you with entering text, equations, *etc.*, in your abstract. **NB** Abstracts are supposed to stand by themselves, this means no footnotes, no cross-references, no figures, no tables, *etc.*

I suggest avoiding the use of the defined acronyms in abstracts *i.e.*, spell them out rather than using the glossary commands. This is due to the fact that the `glossaries` package (that is being used to support acronyms) does not directly provide support for multiple languages and because I do not understand how to programmatically create plurals of acronyms in Swedish or other languages. Even in an English abstract, it is desirable to avoid using the glossary commands - as this makes subsequent processing of the abstracts harder - since one has to make sure that the list of acronyms and their definitions are provided to any program that will process this $\text{\LaTeX}$ source code. For this reason, later versions of this template include the `acronyms.tex` file after the metadata for DiVA.

C.5.2 Structure of the abstracts and summaries

The basic $\text{\LaTeX}$ structure for an abstract or summary is shown below (for the case of an English abstract and a Swedish summary *i.e.*, *sammanfattning*):

```
\begin{abstract}
  \markboth{\abstractname}{}
\begin{scontents} [store-env=lang]
eng
\end{scontents}

\begin{scontents} [store-env=abstracts, print-env=true]
here is where you abstract goes.
\end{scontents}

\subsection*{Keywords}
\begin{scontents} [store-env=keywords, print-env=true]
% If you set the EnglishKeywords earlier, you can
  ↪ retrieve them with:
```

```

\InsertKeywords{english}
% If you did not set the EnglishKeywords earlier then
  ↳ simply enter the comma separate keywords here:
%such as: Canvas Learning Management System, Docker
  ↳ containers, Performance tuning
\end{scontents}
\end{abstract}

\cleardoublepage
\babelpolyLangStart{swedish}
\begin{abstract}
  \markboth{\abstractname}{}
\begin{scontents}[store-env=lang]
swe
\end{scontents}
\begin{scontents}[store-env=abstracts,print-env=true]
Swedish summary goes here
\end{scontents}
\subsection*{Nyckelord}
\begin{scontents}[store-env=keywords,print-env=true]
% SwedishKeywords were set earlier, hence we can use
  ↳ alternative 2
\InsertKeywords{swedish}
\end{scontents}
\end{abstract}
\babelpolyLangStop{swedish}

```

It is important to note that the contents of the `scontents` environment for the abstracts are stored **verbatim**, *i.e.*, the $\text{\LaTeX}$ is **not** executed. The reason for this is to be able to later have a program that can manipulate the source $\text{\LaTeX}$ to convert it to HTML for use in announcements, calendar events, and for DiVA. This means that if you write the following:

```

\begin{scontents}[store-env=abstracts,print-env=true]
\input{abstract.txt}
\end{scontents}

```

what will end up in your abstract in the metadata save for DiVA will simply be: "`\inputabstract.tex`" – which means that someone will have to cut and paste your actual abstract to insert it into DiVA.

It is also important to that that the following lines:

```

\begin{scontents}[store-env=lang]
eng

```

```
\end{scontents}
```

must to be before the `scontents` environment for the abstracts and keywords – as these lines indicate what language the subsequent abstract and keywords are in. The three-character code used for the language is the ISO 639-2 Code – specifically the "B" (bibliographic) variant of these codes — as these codes are used in the DiVA metadata to tag what language is used.

C.5.3 Acronyms

You may want to define an acronym to help you with your writing, as this can both reduce the amount of typing and help your reader by providing consistent use of acronyms. The acronyms' definitions can be found in the file *lib/acronyms.tex*. The file contains some examples. I generally try to sort the lines to help find which acronyms I already have defined and keep track of the new one(s) I want to add.

C.5.4 Some predefined macros to help when writing

The file *lib/defines.tex* includes some macros that will help you when writing. This includes `\etc`, to give you “*etc.*,” `\eg`, `\ie`, and `\etal`. The file also defines `\first`, `\Second`, ... `\eighth` to give you *(i)*, *(ii)*, *(iii)*, ... *(viii)*. Note that ‘*Second*’ is written with an initial capital letter to avoid conflict with the unit ‘second’ in the `siunitx` package.

C.5.5 Additional abstract(s)

All theses at KTH are **required** to have an abstract in both *English* and *Swedish*. However, in addition to this, many students want to add abstracts in additional languages. The template comes pre-configured with places for abstracts in several other languages. If there is a language that you want to use that is not already supported, there are directions for how to add an additional language. If there are abstracts in languages that you do not want, please delete them or comment them out (see Appendix C.5.6).

C.5.6 Removing and hiding parts that you do not want

It is quite likely that you will find parts of the template that you do not want/need. One way of dealing with this is to delete them, and another way is to comment them out. Personally, I like to comment things out, in case I

actually do want to be able to read it in the L^AT_EX file or uncomment it later. To comment out a portion of the file, simply use the following environment:

```
\begin{comment}
  *** what you want to comment out ***
\end{comment}
```

For example, if you are not interested in the Swedish language `todo` notes, you can look for lines with “\sweExpl” in them and comment them out (or delete them).

C.5.7 Removing the README_notes

At some point you will no longer want this README information. You can remove it by removing the line `\include{README_notes/README_notes}` – from the *examplethesis.tex* file. You can then remove the **README_notes** directory.

Unless you are an examiner or an administrator you can delete the file: `README_notes/README_examiner_notes.tex` and delete the include of this file from near the end of the template (*i.e.*, *examplethesis.tex*). You can also delete the directory **README_notes/README_examiner-figures**.

C.6 Copyright or Creative Commons License

It is possible to have several variants of the bookinfo page*:

copyright If you want to have a bookinfo page, include the line saying `\bookinfo` page.

Creative Commons (CC) If you want to have a bookinfo page but want to have a Creative Commons license, then include `\bookinfo` page and use and configure the `doclicense` package as described below.

none If you do **not** want to have a bookinfo page, comment the line saying `\bookinfo` page and add a `\cleardoublepage`.

*When printed double sided, the bookinfo page is the back of the title page.

For background about Creative Commons licenses, see: <https://www.kb.se/samverkan-och-utveckling/oppen-tillgang-och-bibsamkonsortiet/open-access-and-bibsam-consortium/open-access/creative-commons-faq-for-researchers.html> and <https://kib.ki.se/en/publish-analyse/publish-your-article-open-access/open-licence-your-publication-cc>.

Note that the lowercase version of the Creative Commons license has to be used in the modifier, *i.e.*, one of: by, by-nc, by-nd, by-nc-nd, by-sa, by-nc-sa, or zero. For the list of supported licenses, see the documentation for the `doclicense` package.

Note that if the `doclicense` package is used, it automatically redefines `\bookinfo` to be `\bookinfoCC`.

C.6.1 Example configuration to have a CC BY-NC-ND license

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
  version={4.0},
  hyphenation={RaggedRight},
]{doclicense}
```

Note that the option “hyphenation=RaggedRight” can be used with the configuration of the package to set the license information with a ragged right margin rather than as a filled and justified paragraph.

C.6.2 Example configuration to have a CC BY-NC-ND license with a Euro symbol rather than a Dollar sign

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
  version={4.0},
  imagemodifier={-eu-88x31}, % to get Euro symbol rather
  ↪ than Dollar sign
  hyphenation={RaggedRight},
```

```
] {doclicense}
```

C.6.3 Example configuration to have a CC0 license

```
\usepackage [
  type={CC},
  modifier={zero},
  version={1.0},
] {doclicense}
```

C.7 Use of fonts within the thesis

The choice of fonts is a very individual matter and may be affected by the kind of content that you are trying to write, the language that you are writing in, and what you want to convey to your reader. However, some points to keep in mind are:

- Use fonts with serifs for the body of your thesis, their presence makes it much easier for your reader.
- Use sans serif fonts for headings. This helps your reader distinguish them from the body.
- Be very careful when using fonts that are not widely available*. Unless you embed the fonts that you have used, your reader may not see what you want them to see. Ideally, you should embed all fonts – even if you only embed the subset you use.
- Although there are fonts that have a huge number of characters in them, they might not have the characters that you need.
- There are also fonts that, although they have a vast number of characters in them, do not have the math table that L^AT_EX needs to be able to set mathematical content[†].

*For example, even though it is widely used, not everyone has the Arial font.

[†]An example of such a font is Google's Noto font. Even though it includes a vast number of characters, it lacks a math table – although there is an awareness of this missing feature

What can you do when the fonts you use are missing characters that you need to use? One solution is to use a font that has the character(s) that you want and then make use of them in the places that you need to.

The details of working with different fonts and characters is a rather complex area and not for the faint-hearted. However, if you **really** want to have specific characters, X_YLaTeX and LuaLaTeX have the means to help you realize what you want.

C.8 One big thesis file or a master file with includes of the parts

While many students split their thesis into multiple files (such as introduction.tex, background.tex, method.tex, what-you-did.tex, results-and-analysis.tex, discussion.tex, conclusion-and-future-work.tex) and then include these in their main document (with a series of \includexxxx), my experience is that it actually makes it hard to be consistent in the thesis. For example, you cannot do a simple global replacement when you have decided to introduce a particular acronym. It also makes searching for things difficult, as Overleaf's search function only works on individual files*. Personally, I find it hard to correct LaTeX errors when the file is split in this way - since Overleaf does not make it simple to find the root cause of a problem when the chapters are included in this way. There can be a problem with compiling the project in Overleaf, as Overleaf does not always handle the separate files as one document (unless you use the functions to tell LaTeX that a file is part of a larger document and identify the parent document). Although I have had some students do this splitting successfully and they liked being able to compile just a part of their report; I've personally had strange errors occur with it - hence I did not use this with the template.

There are some advantages to splitting the document into different parts:

- Overleaf has a limit on the number of changes that it can track - but this limit is per file! [Yes, I have gotten bitten by this when I have put in more changes and comments than the limit and had to stop marking up a manuscript.]
- Additionally, Overleaf has a per file size limit (*i.e.*, how large a file can

*Note that this is not a problem if you use emacs and a tags file, as this can do searching over the whole set of files and even a tags based query and replace.

be) - again this is per file [Yes, I have gotten bitten by this when exporting a Jupyter notebook that produced a LaTeX file larger than 50 MB.]

- This is useful when different students (in a 1st cycle degree project) are writing different parts of the report (in this case the divisions can be even at the section or subsection level).

Similarly, many students like to group their figures along with their chapters, *i.e.*, introducing a folder for each part of the report and placing both the text and the figures relevant to this section into the relevant folder. A similar approach can be used with included code snippets, tables, *etc.*

Ultimately, I think the main issue is the degree to which the separate files are separate and can be worked on as if they were very independent. This generally is true in third-cycle theses, as the chapters tend to be rather independent - typically with one conference/journal paper as the focus of a chapter. However, my experience is that first-cycle theses have very highly interdependent parts, while 2nd cycle theses are split between highly interdependent and highly independent.

However, some might find the question of splitting or not to be a matter of taste or perhaps different ways to approach organizing their writing. So you and your supervisor(s) and examiners might want to discuss what choice is most suitable for your purposes.

Appendix D

README and notes about the template

This document, written by Gerald Q. Maguire Jr, describes the thesis template that I have developed for use at **KTH Royal Institute of Technology (KTH)** and provides some background about why it is the way that it is. It is important to note that the template is **not prescriptive**, as not every thesis will have all the parts the template shows. However, if there is something that you decide to leave out, you should make a conscious decision to do so, and you should consider the impact this may have on your thesis being approved by the examiner.

Fundamental to the design of the template are several key factors:

- Helping students be successful in their degree project,
- Helping students produce a high-quality thesis, and
- Supporting all of the (relevant) phases of the degree project process.

Several thousand theses are written each year by **KTH** students. Every approved thesis will be entered into **Digitala Vetenskapliga Arkivet (DiVA)** (independent of whether the full text is made available via **DiVA**). Collecting the data necessary for **DiVA** was a major driving force in the design of the template. This data is useful for many of the phases of the degree project, such as announcing the oral presentation.

This template is **not** designed for use by **TIMTM** and **TMMTM** students - as students in these two programs are using a different structure for their reports (there is another template available for them).

This document is a work in progress.

D.1 Introduction

This template evolved (radically) from an earlier thesis template that was widely used at **KTH**. The direction of this evolution was based on the DOCX template developed over many years for use with students for whom I was the examiner and/or supervisor. The suggested structure and contents of the thesis reflect my experience as an examiner for more than 590 degree projects and the experience I have had as a teacher and examiner for the course II2202 Research Methodology and Scientific Writing. The template also reflects my interest as a member of KTH's Language Committee in facilitating the parallel use of English and Swedish at **KTH**, as well as supporting other languages. The latter aspect reflects my experience with double-degree students, who often need to have at least the abstract of their thesis in their home university's language(s). The thesis template also reflects my experience in entering the metadata for hundreds of theses into **DiVA** and announcing a very large number of degree project seminars.

Appendix **D.4** describes several different groups of users and how the template is relevant to them.

Several major thoughts were influenced the design of this template:

- Thought 1** The template should help a student be successful in their degree project and help them produce a high-quality thesis in conjunction with their degree project.
- Thought 2** The template should help support all of the (relevant) phases of the degree project process.
- Thought 3** Redundant data entry should be minimized to increase consistency.
- Thought 4** There are several thousand theses written each year at **KTH**. Theses are the second most common type of publication at **KTH**.
- Thought 5** Every approved thesis will have at least its metadata entered into **DiVA**. **DiVA** features multi-language support for title, subtitle, abstract, and keywords.

D.2 Delimitations

This template is **not** designed for use by **TIMTM** and Media Management **TMMTM** students - as students in these two programs are using a different

structure for their reports (there is another template available for them).

Additionally, I have been told by one of my colleagues in applied mathematics that theses in this area generally do not follow the **Introduction, Methods, Results, and Discussion (IMRAD)** structure.

Some parts of the template are conditional based on the value of a switch: `\ifinswedish`. The idea is to easily have a single template that supports theses written in English or Swedish. However, in many places, the conditional has not been used but could be. Examples of this include the Swedish names for chapters and sections. Generally, this information is in a note after the English chapter or section name. More complete implementation of the use of this condition remains as future work.

The template does not fully support the G5 paper format. In particular, the **KTH** cover (produced with `\kthcover`) and back cover (produced with `\kthbackcover`) have only been adapted for A4 paper. Support for G5 paper remains as future work.

The handling of the subject area (Swedish: Område för examensarbete) is currently incomplete and remains as future work. Personally, I'm still struggling to understand the rules and how one knows what the correct values are (especially for cases of *(i)* dual degrees and *(ii)* combinations of technical subjects and education degrees).

D.3 Structure of the files for the template

Table **D.1** shows the structure of the files for the template. These files are generally taken from an existing Overleaf project, a ZIP file, or a github.

One hope is that by automatically extracting information from various sources, this information is more likely to be *correct* and *consistent* (supporting **Thought 3**). This approach has been used to generate two of the files used for the template. These files are:

1. The file `custom_configuration.tex` contains macros and values for configuring a project. These values are generally expected to be known at the start of the project, *e.g.*, author(s), supervisor(s), examiner, course code for the degree project program code, *etc.* While this file can be manually edited, it was designed to be generated by a program that I have written that extracts most of the data from the Canvas course being used in conjunction with the degree project. One of the goals of using such a program is to extract data from Canvas automatically, the **KTH** profile **Application Programming Interface (API)**, **Kurs- och**

programplaneringssystemet (KOPPS), and other sources. The macros for defining this information are described in Sections **C.4**, **D.5.1**, and **D.5.2** - for authors, supervisors, and examiner (respectively).

2. The file `schools_and_programs.ins` contains the English and Swedish names of schools and programs. A program extracted this information from **KOPPS**.

We will assume that these files have been generated by someone. Later we will examine who this someone might be for each of these files.

Table D.1: Structure of files for the template

bibstyle	directory containing files related to the style of the bibliography	
	<code>myIEEEtran.bst</code>	a bibtex style file
figures	directory containing files for figures	
lib	directory containing various library files	
	<code>acronyms.tex</code>	a place to define the acronyms that will or might be used
	<code>defines.tex</code>	some generally useful defines
	<code>includes-after-hyperref.tex</code>	a special include file for packages that have to be included after the hyperref package
	<code>includes.tex</code>	a centralized place to include packages that might be useful
	<code>kthcolors.tex</code>	defines a number of colors from the KTH palette
	<code>pdf_related_includes.tex</code>	includes to be able to add the title and other information to the PDF file
	<code>schools_and_programs.ins</code>	English and Swedish names of schools and the programs
<code>custom_configuration.tex</code>	macros and values for configuring a project	
<code>examplethesis.tex</code>	an example of the thesis itself	
<code>kth_logo.png</code>	the KTH logo for use on the cover	
<code>KTH_ROYAL_INSTITUTE_OF_TECHNOLOGY_logotype.png</code>	KTH logotype for use on the English language cover	
<code>kththesis.cls</code>	the kththesis class file	
<code>README_notes.tex</code>	these notes	
<code>references.bib</code>	references that may be cited in the thesis	

D.4 Expected users and their differences

This template is relevant to several different sets of users:

- Users 1** Author or Authors (see Appendix [C.1](#)),
- Users 2** Those working together with the author(s) during the degree project process (see Appendix [D.5](#)),
- Users 3** Administrative staff working with the document after it has been approved by the examiner (see Appendix [D.6](#)), and
- Users 4** The (hopefully) many (human) readers of the final document (see Appendix [D.7](#)).
- Users 5** The (hopefully) many computers reading the metadata and the full text of the final document (see Appendix [D.7.1](#)).
- Users 6** Those who are maintaining or updating this template (see Appendix [D.7.2](#)).

Each of these different sets of users has different needs and perspectives. The following subsections describe these needs and perspectives.

For information for authors, see Appendix [C](#) - located in the file `README_author.tex`.

D.5 Those working in parallel with the authors(s) during the degree project

Those working together with the author(s) during the degree project process include the examiner, supervisor(s), and the opponent(s).

D.5.1 Supervisor

If a degree project is done in industry, there is generally an industrial supervisor in addition to the academic supervisor(s). The template supports up to 3 supervisors (typically an academic supervisor, an industrial supervisor, and sometimes an additional academic or industrial supervisor). The choice of up to three reflects my experience and observation of prior theses in [DiVA](#). Note that there is expected to be at least one supervisor. The supervisors are enumerated as A, B, and C. For each of A, B, and C as appropriate, replace the "X" in the following macros:

<code>\supervisorXsLastname{ }</code>	the last name of the supervisor
<code>\supervisorXsFirstname{ }</code>	the first name of the supervisor
<code>\supervisorXsEmail{ }</code>	e-mail address of the supervisor

If the supervisor is from within **KTH**, then add their KTHID, School, and Department info:

<code>\supervisorXsKTHID{ }</code>	the supervisor's kthid
<code>\supervisorXsSchool{ }</code>	the school of the supervisor
<code>\supervisorXsDepartment{ }</code>	the department of the supervisor

If the supervisor is from outside of **KTH**, then add their organization with:

<code>\supervisorXsOrganization{ }</code>	the supervisor's organization
---	-------------------------------

D.5.2 Examiner

I assume that there is only a single examiner for a given thesis*. For this examiner, the relevant macros are:

<code>\examinersLastname{ }</code>	the last name of the examiner
<code>\examinersFirstname{ }</code>	the first name of the examiner
<code>\examinersEmail{ }</code>	e-mail address of the examiner

If the examiner is from within **KTH**, then add their KTHID, School, and Department info:

<code>\examinersKTHID{ }</code>	the examiner's kthid
<code>\examinersSchool{ }</code>	the school of the examiner
<code>\examinersDepartment{ }</code>	the department of the examiner

*Statistically, there are very few theses with multiple examiners, and this generally occurs for students either in a double degree program or when there are two students in a 1st cycle degree project from different schools, then there might be one examiner for each student. As the case of more than one examiner occurs very infrequently, I have left it for future work. The pseudo-JSON structure is set up to handle multiple examiners, but additional macros would be needed in a similar fashion as used for multiple supervisors, and this metadata would have to be conditionally added where appropriate.

If the examiner is from outside of **KTH**, then add their organization with:

`\examinersOrganization{}` the examiner's organization

I assume that someone (such as the examiner) will generate the file: `custom_configuration.tex`. This assumption is based upon the fact that the examiner knows who the student or students are who will be working on a given degree project, who the supervisor or supervisors are, what program the student is in, course code, Ideally, this file should be generated automatically by some computer program so that each student or pair of students in a group gets a customized template automatically via the Canvas course. However, currently, the file is generated using a command line program (`create_customized_JSON_file.py`) to generate a **JavaScript Object Notation (JSON)** file. Subsequently, a separate program (`customize_LaTeX_project.py`) takes this **JSON** data and creates the appropriate **L^AT_EX** commands and inserts this information into the file and then inserts this file into a ZIP file, either replacing or augmenting the `custom_configuration.tex` within this ZIP file (if one exists). There is an option for this second program `-initialize` that causes the program to simply replace the file rather than appending the new information to the end of the file.

The above programs are available from <https://github.com/gqmaguirejr/E-learning>. The README file for this GitHub contains information about how to run the programs, their options, and gives examples.

D.5.3 Opponent(s) and oral presentation

Unlike the supervisors and examiner, the macros related to the opponent and oral presentation are in the `examplethesis.tex` file. The macro for the opponent(s) is:

`\opponentsNames{}` the names (in normal name order) of the opponent or opponents

When there are multiple opponents, separate their names with `'\&'`; for example, A. B. Normal `\&` A. X. E. Normalè.

For the oral presentation, the following macros are filled in once the examiner has scheduled your oral presentation:

`\presentationDateAndTimeISO{}` date and time of the presentation is ISO format, for example:
2022-03-15 13:00

<code>\presentationLanguage{ }</code>	three letter abbreviation for the language of the presentation according to three letter ISO 639-2 Code – specifically the "B" (bibliographic) variant of these codes (note that this is the same language code used in DiVA), generally eng or swe
<code>\presentationRoom{ }</code>	a room name and/or "via Zoom https://kth-se.zoom.us/j/aaaaaaaaaaaa "
<code>\presentationAddress{ }</code>	location of the room, for example: Isafjordsgatan 22 (Kistagången 16)
<code>\presentationCity{ }</code>	city where the presentation occurs, generally: Stockholm

D.6 Administrative staff

Once a thesis is approved by the examiner we need to add the TRITA number. The TRITA number is assigned by the student affairs office of the school from an annual series of numbers.

D.6.1 What is a TRITA number and why does each approved thesis get assigned one?

TRITA stands for Transactions for the Royal Institute of Technology, with the letter "A" appended to it. The TRITA definition is the 1971 report, "Mall för publikationsserier vid Kungl. Tekniska högskolan i Stockholm"", TRITA-LIB-1001, <http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-127656>.

The format for TRITA numbers for degree projects is TRITA-(school acronym)-EX-YYYY:nnnn, where nnnn is a sequential number starting from 1 each year with the numbers assigned in chronological order to approved theses ("numren delas ut kronologiskt först när examinatorn godkänt arbetet." - according to one of KTH's archivists). Note that the list of assigned TRITA numbers is archived each year*. The year, YYYY, is based on the year that

*It seems that this archiving is done twice a year.

the thesis was *approved*.

The TRITA number value can be set with a macro that takes two arguments: series and year:number as shown below:

```
% for entering the TRITA number for a thesis
\trita{TRITA-EECS-EX}{2022:00}
```

D.6.2 Where does the TRITA number go?

The TRITA number will appear on the back cover of the thesis. It is also stored as part of the metadata entered into DiVA.

D.6.3 What does this mean in practice?

Currently, at EECS the TRITA number is only assigned to the thesis when the examiner has approved the thesis and submitted the PDF of the approved thesis (with cover) to the student affairs office. Of course, this does not make much sense because the back cover is already on the thesis! This means that someone in the student affairs office must either (i) edit the sequential number part of the TRITA number (using some PDF tool) or (ii) they need to make a new back cover and replace the existing back cover. A better solution would be to inform the examiner of the TRITA number and the examiner can see that this number is inserted into the macro shown above and this can enable the number to appear on the back cover and as an added bonus be included in the metadata for DiVA.

Note that it is expected that in 2023, this process will change – thus the assignment of the TRITA number and the application of the back cover would be done by the student affairs office (as only they have the relevant information)*.

D.6.4 Entering the metadata into DiVA

If a thesis has used this template the “For DIVA” page contains the metadata for DiVA and an administrator can cut and paste this data into DiVA. Alternatively, this metadata can be extracted with a program from the PDF file to produce a JSON file that can subsequently be used to create a MODS file for import into DiVA. The L^AT_EX compiler can in many cases produce a file called “fordiva.json” that contains the metadata.

*Note to maintainers: This means that the back cover can be removed from this template.

The programs that can be used to extract data and to take a **JSON** file and create a MODS file are available from <https://github.com/gqmaguirejr/E-learning>.

Note that the import of the MODS file does **not import the collaboration data**, even though this is in the file. This is a limitation of the **DiVA** import function. Therefore, this information has to be manually entered along with uploading of the PDF file itself.

D.7 (Human) Readers of the thesis

Some theses have very few downloads from **DiVA**, while some have had hundreds of thousands of downloads. Therefore, you should remember that you have a wide range of human readers of your thesis. The readers include other students looking for information related to their own thesis or because they are interested in the future work that you have suggested to work on for their own degree project. Additionally, researchers who are looking for your results may find your thesis relevant to them. In many cases, companies will look at theses for ideas about what the state of the art is - in several cases, theses have been important as “prior art” and this invalidated patents that had been issued if the patent was submitted after the thesis became public (hence it pays to get theses public as soon as possible). Other human readers are the UKÄ review teams that examine the degree programs offered at **KTH**. Finally, as **KTH** is a public agency, it is important that the general public know what is done at **KTH***.

D.7.1 Machines reading the metadata or full text of the thesis

The file `pdf_related_includes.tex` contains $\text{\LaTeX}$ code that stores the title, author(s), and keyword information into the PDF document in such a way that if you ask for the properties of the PDF file you will get this data. This information makes it easier for machines to get this information from the PDF file.

Additionally, many search engines (such as Google’s search engine) mine **DiVA** for the metadata and if the full text of the thesis is published via **DiVA** then they also process the full text of the thesis. The result is that search engines can find the content in these theses. This is likely to increase the

*This is an important part of the Swedish Offentlighetsprincipen.

probability that someone will download your thesis if they think it is relevant to them – increasing the number of your human readers (see Appendix D.7).

D.7.2 Template author and maintainers

KTH periodically changes the cover design for theses, introduces new programs of study, eliminates programs of study, reorganizes administratively, and faculty move between schools, departments, and divisions. It can be expected that this template will need to evolve with these changes.

For example, if there is a change in schools or programs then there needs to be changes made to the file `schools_and_programs.ins`. While the current file was extracted from KOPPS, the program that does this will need to be replaced because further development of KOPPS has been terminated by KTH’s central IT unit which plans to transition all of this information to Lokalt adb-baserat dokumentationssystem (LADOK).

As another example, on 13 December 2021 there was a change in the KTH cover for 1st and 2nd theses, and the cover generator web service was shutdown. The initial draft version of the cover used a proprietary font (TheSans B4 SemiLight and TheSans B6 SemiBold). The version that was publicly introduced uses another proprietary font (Arial) and officially only existed as a DOCX file for a thesis in Swedish. The result is that I had to make my own version in L^AT_EX to try to emulate the DOCX cover. This led to a lot of effort, but one can get a reasonable cover with the correct font as described in Appendix C.2.

D.8 While writing

As was noted in Appendix C.1, the thesis template contains lots of examples, notes, and comments. One method to provide additional information is the use of `\todo`. Several different types of `\todo` notes have been used in the thesis. These are described in Appendix D.8.1.

D.8.1 Conventions for `\todo` notes

The example thesis text includes extensive comments, directions, and warnings. These follow the form shown below:

```
\generalExpl{Comments/directions/... in English}
\sweExpl{Text på svenska}
\engExpl{English descriptions about formatting}
```

```
\warningExpl{warning}
```

and appear as:

Comments/directions/... in English
Text på svenska
English descriptions about formatting
warning

Each of the above is a macro, so as usual in $\text{\LaTeX}$ you can redefine it - even defining it to produce nothing! Several previous students have placed these re-definitions in the `custom_configuration.tex` file.

D.8.2 Turning on and off the README_notes

As the various README notes are targeted at different readers, you may or not want to see them. It is very easy to turn them on or off by adding or removing a percent (`'%`) character before the relevant `\begin{comment}` and `\end{comment}` comments around each set of notes.

For example, if you are a student writing a thesis, I suggest turning off everything except for the `README_author.tex` and `README_notes.tex` sets of notes. However, I would suggest keeping the other README files around (at least for a little while) as a source of examples of how to do things. Despite having spent a very large number of hours working on the template and drafts of students' theses, I find some of the README files very helpful as a reminder of how to do things.

D.8.3 Removing the README_notes

At some point you will no longer want this README information. You can remove it by removing the line `\include{README_notes/README_notes}` - from the `examplethesis.tex` file. If you have removed the other README* files from the **README_notes** directory, you can then remove the **README_notes** directory.

D.8.4 Removing the README_notes

At some point, you will no longer want this README information. You can remove it by removing the line `\include{README_notes/README_notes}` - from the `examplethesis.tex` file. If you have removed the other

README* files from the **README_notes** directory, you can then remove the **README_notes** directory.

D.8.5 Removing unused fonts

This version of the template may also have some font information, in the form of OpenType Font files (with the extension “.otf”) and TrueType Font font files (with the extension “.ttf”). If you are not using these fonts (and no longer are using any of the README files), then you can delete these font files.

README acronyms

API	Application Programming Interface
DiVA	Digitala Vetenskapliga Arkivet
IMRAD	Introduction, Methods, Results, and Discussion
JSON	JavaScript Object Notation
KOPPS	Kurs- och programplaneringssystemet
KTH	KTH Royal Institute of Technology
LADOK	Lokalt adb-baserat dokumentationssystem
TIMTM	Interactive Media Technology first
TMMTM	Media Management first

€€€€ For DIVA €€€€

```
{
  "Author1": { "Last name": "Eriksson",
    "First name": "Arvid",
    "Local User Id": "u100001",
    "E-mail": "arveri@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    }
  },
  "Cycle": "2",
  "Course code": "DA231X",
  "Credits": "30.0",
  "Degree1": { "Educational program": "Degree Programme in Computer Science and Engineering",
    "programcode": "CDATE",
    "Degree": "Degree of Master of Science in Engineering",
    "subjectArea": "Computer Science and Engineering"
  },
  "Title": {
    "Main title": "Adversarial training for the automatic diagnosis of the electrocardiogram",
    "Subtitle": "Making deep learning analysis of the electrocardiogram robust",
    "Language": "eng" },
    "Alternative title": {
      "Main title": "Motstridig träning för automatiskt diagnostisering av elektrokardiogram",
      "Subtitle": "Ett sätt att göra djupinlärningsanalyser av elektrokardiogram robusta",
      "Language": "swe"
    },
  },
  "Supervisor1": { "Last name": "H. Ribeiro",
    "First name": "Antonio",
    "Local User Id": "u100003",
    "E-mail": "anhr@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
      "L2": "Computer Science" }
  },
  "Supervisor2": { "Last name": "Jaldén",
    "First name": "Joakim",
    "Local User Id": "u100003",
    "E-mail": "jalden@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    }
  },
  "Examiner1": { "Last name": "Fransén",
    "First name": "Erik",
    "Local User Id": "u1d13i2c",
    "E-mail": "erikf@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
      "L2": "Computer Science" }
  },
  "Cooperation": { "Partner_name": "Uppsala University",
    "National Subject Categories": "10201, 10203",
    "Other information": { "Year": "2026", "Number of pages": "xxl,75"},
    "Copyrightleft": "copyright",
    "Series": { "Title of series": "TRITA-EECS-EX", "No. in series": "2023:0000" },
    "Opponents": { "Name": "A. B. Normal & A. X. E. Normalè",
      "Presentation": { "Date": "2022-03-15 13:00"
        "Language": "eng"
      },
      "Room": "via Zoom https://kth-se.zoom.us/j/ddddddddddd"
    },
    "Address": "Isafjordsgatan 22 (Kistagången 16)"
    "City": "Stockholm" },
    "Number of lang instances": "2",
    "Abstract[eng ]": €€€€

\generalExpl{Enter your abstract here!}
Write an abstract that is about 250 and 350 words (1/2 A4-page) with the following components:
% key parts of the abstract
\begin{itemize}
  \item What is the topic area? (optional) Introduces the subject area for the project.
  \item Short problem statement
  \item Why was this problem worth a Bachelor's/'Masters thesis project? (\ie, why is the problem both significant and of a suitable degree of difficulty for a Bachelor's/'Masters thesis project? Why has no one else solved it yet?)
  \item How did you solve the problem? What was your method/insight?
  \item Results/Conclusions/Consequences/Impact: What are your key results/\linebreak[4]conclusions? What will others do based on your results? What can be done now that you have finished - that could not be done before your thesis project was completed?
\end{itemize}

€€€€,
"Keywords[eng ]": €€€€
```

Deep learning, Adversarial training, Electrocardiogram €€€€,
"Abstract[swe]": €€€€

```
\generalExpl{Enter your Swedish abstract or summary here!}  
\sweExpl{Alla avhandlingar vid KTH \textbf{måste ha} ett abstrakt på både \textit{engelska} och  
\textit{svenska}.\\  
Om du skriver din avhandling på svenska ska detta göras först (och placera det som det första  
abstraktet) - och du bör revidera det vid behov.)  
  
\engExpl{If you are writing your thesis in English, you can leave this until the draft version that  
goes to your opponent for the written opposition. In this way, you can provide the English and  
Swedish abstract/summary information that can be used in the announcement for your oral  
presentation.\\If you are writing your thesis in English, then this section can be a summary targeted  
at a more general reader. However, if you are writing your thesis in Swedish, then the reverse is  
true - your abstract should be for your target audience, while an English summary can be written  
targeted at a more general audience.\\This means that the English abstract and Swedish sammnfattning  
or Swedish abstract and English summary need not be literal translations of each other.)  
  
\warningExpl{Do not use the \textbackslash glspl{\} command in an abstract that is not in English,  
as my programs do not know how to generate plurals in other languages. Instead, you will need to  
spell these terms out or give the proper plural form. In fact, it is a good idea not to use the  
glossary commands at all in an abstract/summary in a language other than the language used in the  
\texttt{acronyms.tex file} - since the glossary package does \textbf{not} support use of more than  
one language.)  
  
\engExpl{The abstract in the language used for the thesis should be the first abstract, while the  
Summary/Sammanfattning in the other language can follow}  
  
€€€€,  
"Keywords[swe]": €€€€  
Djupinläring, Motstridig träning, Elektrokardiogram €€€€,  
}
```

acronyms.tex

```
%%% Local Variables:
%%% mode: latex
%%% TeX-master: t
%%% End:
% The following command is used with glossaries-extra
\setabbreviationstyle[acronym]{long-short}
% The form of the entries in this file is \newacronym[label]{acronym}{phrase}
% or \newacronym[options]{label}{acronym}{phrase}
% see "User Manual for glossaries.sty" for the details about the options, one example is shown below
% note the specification of the long form plural in the line below
\newacronym[longplural={Debugging Information Entities}]{DIE}{DIE}{Debugging Information Entity}
%
% The following example also uses options
\newacronym[shortplural={OSes}, firstplural={operating systems (OSes)}]{OS}{OS}{operating system}

\newacronym{ECG}{ECG}{electrocardiogram}

% note the use of a non-breaking dash in long text for the following acronym
\newacronym{IQL}{IQL}{Independent -QLearning}

% example of putting in a trademark on first expansion
\newacronym[first={NVIDIA OpenSHMEM Library (NVSHMEM\texttrademark)}]{NVSHMEM}{NVSHMEM}{NVIDIA OpenSHMEM
↔ Library}

\newacronym{KTH}{KTH}{KTH Royal Institute of Technology}

\newacronym{LAN}{LAN}{Local Area Network}
\newacronym{VM}{VM}{virtual machine}
% note the use of a non-breaking dash in the following acronym
\newacronym{WiFi}{-WiFi}{Wireless Fidelity}

\newacronym{WLAN}{WLAN}{Wireless Local Area Network}
\newacronym{UN}{UN}{United Nations}
\newacronym{SDG}{SDG}{Sustainable Development Goal}
\newacronym{FGSM}{FGSM}{Fast Gradient Signed Method}
\newacronym{PGD}{PGD}{projected gradient descent}
\newacronym{APGD}{APGD}{Auto-PGD}
\newacronym{CNN}{CNN}{convolutional neural network}
```